



STG-SurviNet: Mining Spatial, Temporal, and Survival Patterns in Urban Infrastructure Resolution Times Across Chicago Community Areas

Syafira Najema Putri Anisa (24031554013), Ketut Shridhara (24031554115), Naura Kanaya Putri Masruri (24031554138)

Moh. Khoridatul Huda, S.Pd., M.Si., Ph.D.

Kelas 2024-INT

Program Studi Sains Data, Universitas Negeri Surabaya

Abstract: This study analyzes infrastructure complaint resolution times across Chicago's 311 system using STG-SurviNet, a hybrid architecture combining a Graph Convolutional Network, a Temporal Convolutional Network, and a Deep Cox Proportional Hazards layer. The study focuses on resolution duration rather than simple completion status, using `duration_days` and event as the survival variables. After preprocessing, the cleaned dataset contains 541,625 valid records and is divided into training, validation, calibration, and testing sets using a 60/20/10/10 split. STG-SurviNet is compared with Kaplan-Meier, Cox Proportional Hazards, and Random Survival Forest using the same 324,975 training records and 54,163 test records. STG-SurviNet achieves the highest C-index of 0.7075, the lowest Integrated Brier Score of 0.1209, and the lowest MAE of 13.68 days. However, all evaluated models produce a D-Calibration p-value of 0.0000, indicating that their survival probability estimates are not well calibrated. The analysis also includes an ablation study, GMM clustering of learned GCN and TCN embeddings, temporal workload profiling, and delay-risk aggregation by community area and complaint category.

Keywords: *Chicago 311, survival analysis, Graph Convolutional Network, Temporal Convolutional Network, infrastructure complaints, service request resolution.*

Chapter 1 Introduction

Chicago, the third-largest city in the United States, operates one of the most well-documented public complaint systems in the world through its 311 Service Requests platform, which has accumulated over 13.9 million reports since its relaunch in December 2018 (City of Chicago, 2026). This study focuses on 647,521 infrastructure-related complaints filed between 2022 and 2026 across four categories: potholes, streetlight outages, tree debris, and traffic signal failures. Despite a resolution rate of 97%, the more operationally relevant question is not whether a complaint will be resolved, but when. The dataset reveals a heavily right-skewed distribution with a mean resolution time of roughly 22 days against a median of only 4 days, with

some complaints remaining open for over 1,500 days. Unresolved infrastructure complaints carry real consequences: damaged roads pose safety risks to motorists, non-functioning streetlights increase crime exposure at night, and prolonged response times erode public trust in city services. This gap also reflects identifiable spatial and temporal patterns, as community areas carrying high historical backlogs tend to resolve new complaints more slowly, and seasonal surges such as post-winter road damage can push departmental queues beyond capacity and compound delays across the entire system.

Conventional regression approaches cannot adequately model this problem for two reasons. First, the 19,547 complaints that were still open at the time of data extraction

have no closed_date, meaning standard regression models must discard them entirely and introduce survivorship bias by learning only from already-resolved cases. Survival analysis addresses this directly by treating open complaints as right-censored observations, allowing them to contribute to hazard estimation even without a known resolution time, a property that is central to deep survival frameworks such as DeepSurv (Katzman et al., 2018) and DeepHit (Lee et al., 2018). Second, regression treats each complaint independently, ignoring both the relational structure between neighbouring areas and the sequential load history that shapes how quickly a given area can respond. Graph Convolutional Networks have demonstrated strong capability in learning from graph-structured data where the state of each node is shaped by its neighbours (Defferrard et al., 2016; Kipf and Welling, 2017), making them well-suited for encoding the spatial pressure that overloaded areas exert on their surrounding community areas. This study therefore proposes STG-SurviNet, a hybrid architecture that integrates a Graph Convolutional Network, a Temporal Convolutional Network, and a Deep Cox Proportional Hazards layer into a single end-to-end model trained with Cox Negative Partial Log-Likelihood loss. The study is guided by four research questions: first, how the spatial proximity structure between community areas influences resolution duration in neighbouring areas; second, whether the 60-day complaint backlog volume of an area correlates with changes in the resolution hazard of new complaints in that same area; third, whether the proposed GCN + TCN + Deep Cox architecture produces more accurate predictions than classical survival baselines including Kaplan-Meier, Cox PH without graph features, and Random Survival Forest; and fourth, which community areas consistently show low hazard scores and

which spatial-temporal factors best explain those delays. To address these questions, this study aims to construct a spatial graph of Chicago's 77 community areas from official boundary data, build per-area daily complaint volume time series as sequential input to the TCN, train the full architecture end-to-end in PyTorch, and evaluate it against five baselines using C-index, Integrated Brier Score, and D-Calibration. Academically, this work contributes a demonstration of integrating graph learning, sequence modelling, and survival analysis within one jointly trained architecture, an approach that remains underexplored in the urban public service literature, and shows that right-censored open complaints carry informative signals that only survival-based models can properly exploit. Practically, the model produces individual hazard curves that city governments can use to prioritise complaints at high risk of prolonged delay, and spatial hazard maps that support more targeted allocation of field resources before backlogs worsen. Because the required variable structure is identical, the framework is also directly replicable to other city complaint systems such as Jakarta CRM or Surabaya WargaKu once those datasets become publicly available.

Chapter 2 Research Methodology

A. Dataset Description

This study uses the Chicago 311 Service Requests dataset obtained from the official City of [Chicago Open Data Portal](#). The dataset contains public infrastructure complaint records submitted by residents between 2022 and 2026 and is widely used to analyze urban public service performance in the City of Chicago. The original dataset consists of more than 13.9 million records covering various categories of public complaints across the city. In this research, only four infrastructure-related complaint categories were selected as the study subset.

Table 1. Selected Infrastructure complaint categories

Category (sr_type)	Number of Records
Pothole in Street Complaint	327,507
Street Light Out Complaint	180,460
Tree Debris Clean-Up Request	77,202
Traffic Signal Out Complaint	59,385

These categories were selected because they represent infrastructure issues frequently reported by residents and contain clear spatial and temporal information relevant to this study. After the filtering process, the resulting dataset consists of 647,521 complaint records used for further analysis. The dataset also provides several important attributes, including complaint category, complaint submission time, completion time, complaint status, community area identifier, and geographic coordinates. In particular, the variables `created_date` and `closed_date` provide information regarding the submission and completion time of each complaint record, while `community_area`, `latitude`, and `longitude` provide spatial information related to the location of the reported infrastructure issue. In addition, the dataset also includes complaint status information, such as completed and unresolved reports, which reflects the condition of complaint handling within the public service system.

The Chicago 311 dataset was selected because it provides event-level complaint records with detailed temporal and spatial information that includes complaint submission time, completion time, and community area identifiers. These attributes make the dataset suitable for analyzing

infrastructure complaint patterns across different regions and time periods.

B. Research Variable

Table 2. Research Variables

Variable Name	Description
<code>sr_number</code>	Unique identifier for each complaint record.
<code>sr_type</code>	Type of infrastructure complaint submitted by residents, used as a categorical feature in the model.
<code>created_date</code>	Timestamp when the complaint was submitted, used as the start time in survival analysis
<code>closed_date</code>	Timestamp when the complaint was completed. Absent complaints are still Open at the time of data collection.
<code>status</code>	Complaint handling status: Completed, Open, or Canceled. Used as the basis for constructing the event indicator.
<code>Community_area</code>	Community area identifier in Chicago (1-77). Serves as the node ID in the spatial graph component.
<code>ward</code>	Political district identifier in Chicago (1-50), used as an additional feature.

latitude	A geographic coordinate used to verify node assignment in the spatial graph.
longitude	A geographic coordinate is used to verify node assignment in the spatial graph.
Duration _days	Primary response variable representing the number of days from complaint submission to completion or censoring date for Open complaints.
event	Censoring indicator where 1 = Completed (observed event) and 0 = Open (right-censored), meaning the complaint had not yet been resolved at the time of data collection.

C. Data Preprocessing

The preprocessing stage was one of the most important parts of the current progress because the model depends on clean temporal, spatial, and survival variables. First, records with missing community_area, latitude, or longitude. These columns are needed because each complaint must be connected to a community area node in the spatial graph. Without this information, the record cannot contribute properly to the GCN component.

Second, duplicate incidents were removed when the duplicate column was available. This step is important because duplicate complaints can make workload appear larger than it really is. If duplicate reports are not removed, some areas may look more overloaded only because the same incident was reported more than once.

Third, created_date and closed_date were converted into datetime format. For completed complaints, duration_days was calculated as the number of days between closed_date and created_date. For open complaints with missing closed_date, the maximum created_date in the dataset was used as the censoring date. This allows open complaints to remain in the survival analysis instead of being dropped.

Finally, records with negative duration_days were removed because a complaint cannot logically be closed before it is created. After all preprocessing steps, the cleaned dataset contains 541,625 records.

Table 3. Missing Values Summary

Column	Missing	Percentage
community_area	2,960	0.5%
closed_date	8,843	1.4%
Latitude/longitude	1,060	0.2%
Others column	0	0%

D. Graph Convolutional Network (Spatial Components)

A Graph Convolutional Network is a deep learning method that operates directly on graph structures that allow each node to iteratively aggregate information from its neighbors. Different from conventional neural networks that work on structured data such as grids or text sequences, GCNs are designed to capture relational connections between arbitrarily linked entities, making them highly relevant for urban contexts where conditions in one area are influenced by pressures in surrounding areas.

In STG-SurviNet, the GCN represents Chicago's 77 community areas as nodes in a weighted graph $G = (V, E)$ where an edge is formed if two areas are geographically adjacent. Each node v is initialized with a

feature vector x_v that includes the proportion of each complaint category, the average historical resolution time, and the density of complaints per square kilometer. A two-layer GCN then aggregates signals from neighbors so that overload in areas such as Englewood automatically propagates to the embeddings of areas directly bordering it, such as Gage Park and Chicago Lawn. The output of this module is a spatial embedding h_v of dimension d_s per node.

E. Temporal Convolutional Network (Temporal Component)

A Temporal Convolutional Network is an architecture that processes sequential data using dilated causal convolution, allowing the receptive field to grow exponentially without proportionally increasing the network's depth. Compared to LSTMs, TCNs are superior for long time windows because their training can be fully parallelized and they are not prone to the vanishing gradient problem.

For each node v , a time series $S_v = [c_{t-59}, c_{t-58}, \dots, c_t]$ is constructed, where c_t is the number of new complaints received on day t . A 60-day window is selected to capture a full operational cycle of the city department while remaining sensitive to seasonal changes, such as post-winter spikes in road damage. The output of this module is a d_t -dimensional temporal embedding h_t that represents the historical load conditions of the area before the arrival of new complaints.

F. Deep Cox Proportional Hazards (Survival Components)

The Cox Proportional Hazards model captures the relationship between covariates and the hazard function of an event, the instantaneous rate at which an event occurs at time t , given that the event has not yet occurred. Its main advantage is its ability to

naturally handle right-censored observations, so that data for which the event has not yet occurred at the end of the observation period still contribute to the estimation. In the Deep Cox implementation, a neural network layer is used to learn non-linear representations of the covariates before passing them to the hazard function.

The two embeddings are combined via $z_v = \text{concat}([h_v, h_t])$ and then passed to a fully-connected network that produces the log-partial-hazard per observation. The loss function being minimized is the Cox Negative Partial Log-Likelihood, given by $L(\theta) = -\sum_i [\varphi(z_i) - \log(\sum_{j: T_j \geq T_i} \exp(\varphi(z_j)))]$,

which ensures the 19,547 open cases continue to contribute to the hazard estimate. The final output is the individual survival curve $S(t | z_v)$, which shows the probability that a complaint remains unresolved on day t , conditional on the spatial and temporal profile of its origin.

G. Knowledge Discovery Framework and Evaluation

STG-SurviNet is treated as a feature learning tool from which three categories of patterns are extracted after training. The first is spatial patterns: GCN embeddings are clustered using GMM (Gaussian Mixture Model) to identify groups of neighborhoods with similar infrastructure stress profiles, independent of geographic proximity. The question being asked is whether administrative ward boundaries actually reflect service pressure, or whether a different spatial structure emerges from the complaint data itself. The second is temporal patterns: the TCN's learned representations are analyzed to identify which time windows correlate most strongly with resolution delay, with particular attention to post-winter months when road damage complaints spike.

The third is survival risk patterns: Cox hazard scores are aggregated per neighborhood and per complaint category to rank Chicago's 77 community areas by their structural resolution risk, separating neighborhoods where delays are driven by complaint volume from those where delays persist regardless of volume.

Table 4. Evaluation Metrics and Performance Targets

Metric	Description	Target
C-index	Relative risk ranking across observations	≥ 0.75
Integrated Brier Score	Probability accuracy of the survival curve	< 0.20
D-Calibration	Calibration of predicted resolution time distribution	$p > 0.05$
MAE (complete only)	Prediction error in plain days	< 7 days

Model performance is evaluated against three classical survival baselines: Kaplan-Meier as a covariate-free reference, Cox Proportional Hazards using traditional numerical features, and Random Survival Forest as a non-linear tree-based benchmark. All models use the same 324,975 training records and 54,163 test records. Cox PH and Random Survival Forest use a feature matrix constructed from incident-level features, spatial area features, and temporal workload summaries, while Kaplan-Meier estimates a population-level survival function without covariates. Model performance is measured using C-index, Integrated Brier Score, D-Calibration, and Mean Absolute Error. The ablation study also evaluates three reduced STG-SurviNet variants by removing the spatial component, temporal component, or incident-level features. This evaluation design

allows the models and feature groups to be compared using consistent record-level data splits.

Table 5. Baseline Models

Model	Description
Kaplan-Meier	Non-parametric floor, no covariates
Classic Cox PH	Tabular-only parametric benchmark
Random Survival Forest	Non-linear tree-based benchmark

H. Research Workflow

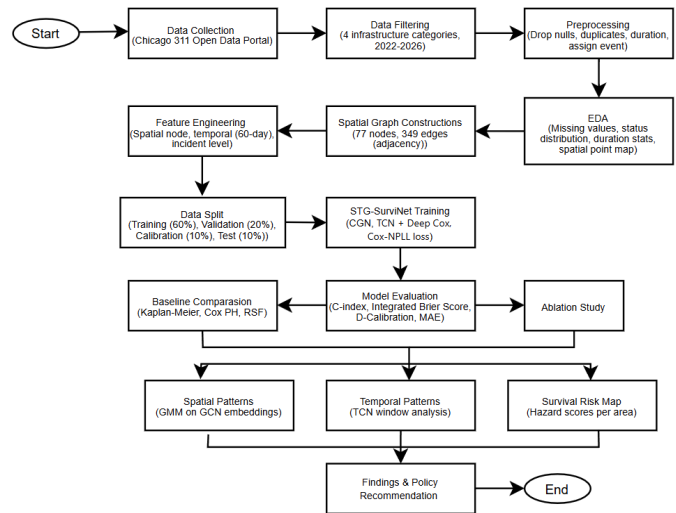


Figure 1. Research Workflow

This study was conducted through a series of systematic stages. Data was collected from the Chicago 311 Open Data Portal, filtered to four infrastructure complaint categories across the 2022–2026 period, yielding an initial dataset of 647,521 records.

The data then underwent preprocessing. Records missing spatial identifiers were dropped, duplicates were removed and also the survival variable `duration_days` was computed for each complaint. Open complaints were retained as right-censored observations using maximum `created_date` as the censoring boundary while Cancelled complaints were

excluded entirely. The cleaned dataset contains 541,625 records.

Exploratory data analysis was then conducted to examine the distribution of complaint statuses, descriptive statistics of resolution duration per category, and spatial point maps confirming geographic structure across Chicago community areas. A spatial graph was subsequently constructed from the official boundary data of Chicago's 77 community areas with 394 weighted edges derived from inverse centroid distances. Model features were organized into three groups: spatial node features per community area, a 60-day daily complaint count sequence per area as temporal input, and incident-level features describing each individual complaint.

STG-SurviNet was trained end-to-end combining a GCN, TCN, and Deep Cox layer minimizing Cox Negative Partial Log-Likelihood loss. Data was split 60/20/10/10 for training, validation, calibration, and testing. Model performance was evaluated using C-index, Integrated Brier Score, D-Calibration, and MAE against three baselines and two ablation variants. The spatial, temporal, and survival risk patterns extracted from the trained model were then synthesized into findings and policy recommendations.

Chapter 3 Results and Discussion

A. Exploratory Data Analysis (EDA) Before Preprocessing

Before preprocessing, initial exploration of the raw dataset was conducted to assess the overall completeness of the data and identify potential issues that needed to be addressed before the data could be used in the modeling pipeline. A missing value analysis was performed across all columns of the dataset, and the results were visualized through the Missing Values Percentage Before Preprocessing bar chart to provide a clearer

picture of the data quality condition prior to any cleaning steps.

Table 6. Missing values count and percentage

Feature	Missing Count	Missing Percentage
closed_date	9721	1.426101
community_area	2950	0.453781
ward	2792	0.429476
latitude	1062	0.163361
longitude	1062	0.163361
parent_sr_number	544249	83.718508
electricity_grid	3569	0.548998
electricity_district	413376	63.587112

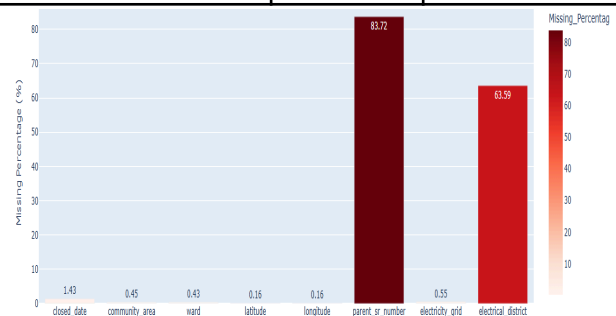


Figure 2. Bar Chart of Missing values

The analysis revealed that most columns had very few missing values. The closed_date column has 9,271 missing values (1.43%), community_area has 2,950 (0.45%), ward has 2,792 (0.42%), and latitude and longitude each have 1,062 (0.16%). However, two columns recorded very high missing values, including parent_sr_number at 83.72% and electrical_district at 63.59%, which clearly dominate the bar chart visualization in dark red. These two columns are not primary variables in the model, so they do not affect the analysis pipeline. Missing values in closed_date are not considered corrupted data but rather indicate that the complaint is still open and will be treated as right-censored observations in the next stage.

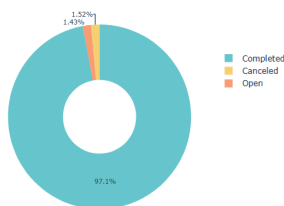


Figure 3. Donut Chart of Raw Tickets Distributions

The next analysis focused on the status column to understand the composition of complaints before filtering. The Raw Ticket Status Distribution donut chart shows that 97.1% of complaints have a status of Completed, while Open accounts for 1.52% and Cancelled for 1.42%.

The dominance of the Completed status confirms that the more relevant research question is not whether a complaint will be resolved, but rather how long it takes to resolve. Open complaints will be retained as right-censored observations, while cancelled complaints will be excluded because they do not represent valid resolutions or survival.

B. Preprocessing

Based on the results of the exploratory data analysis (EDA) conducted earlier, it was found that the raw dataset contained several issues that needed to be addressed before the data could be used in the modeling pipeline, including missing values in critical columns and the presence of duplicate records. Therefore, a series of preprocessing steps were applied sequentially to ensure the consistency, completeness, and compatibility of the data with the survival analysis framework used.

Table 7. Preprocessing Pipeline

Action	Purpose
Drop rows missing community_area, latitude or longitude	Required as spatial identifiers for graph node assignment

Filter duplicate == False	Prevent workload distortion in temporal features
Convert created_date & closed_date to datetime	Enable time arithmetic for duration calculation
Compute duration_days	Primary survival response variable
Fill missing closed_date with max created_date	Right-censoring boundary for open complaints
Assign binary event indicator	Event indicator for Cox model
Remove negative duration_days	Eliminate data entry errors

Rows with empty spatial columns are completely removed because records without a community_area, latitude, or longitude cannot be assigned to any node in the spatial graph and therefore lack usable spatial information. Duplicate records are filtered to avoid inflating the volume of complaints in the time series. The survival variable duration_days is calculated as the integer difference in calendar days between closed_date and created_date. For complaints that were still open at the time of data extraction, the maximum created_date value in the dataset is used as the censoring date, meaning the complaint is known to be unresolved at least up to that point in time. A binary event indicator was then created, with a value of 1 assigned to complaints with a status of Completed and 0 for all other records, including those with a status of Open. Records yielding a negative duration_days value were removed because they indicated data entry errors where the closed_date preceded the created_date. After all cleaning steps were applied, the dataset was reduced to 541,625 valid records.

C. Exploratory Data Analysis (EDA) After Preprocessing

After preprocessing, the cleaned data was grouped by complaint category to calculate descriptive statistics for duration_days. The results show that resolution duration is right-skewed. This means many complaints are resolved quickly, but some complaints remain open or unresolved for a very long time. This pattern supports the use of survival analysis because survival methods are designed to handle time-to-event data with long tails and censored cases.

Table 8. Descriptive Statistics of Resolution Duration by Category

Complaint Category	Count	Mean Duration	Median Duration	75%	Max
Pothole In Street Complaint	243,487	22.43 days	5 days	19 days	1,486 days
Street Light Out Complaint	188,897	15.68 days	2 days	6 days	1,597 days
Traffic Signal Out Complaint	48,857	21.54 days	0 days	1 day	888 days
Tree Debris Clean-Up Request	60,384	13.85 days	5 days	16 days	427 days

Pothole complaints have the largest number of records and an average duration of 22.43 days. Traffic Signal Out complaints have a median duration of 0 days, which means many of them are completed on the same day. Street Light Out complaints have a median of 2 days, but their maximum duration reaches 1,597 days. This shows that even when the typical case is resolved quickly, the category still contains very long-delayed cases.

The boxplot and density plot are useful for showing this pattern visually. The boxplot shows the difference in spread and outliers between complaint categories. The density plot, capped at 60 days, makes the early part

of the distribution easier to compare because the full distribution has very long outliers.

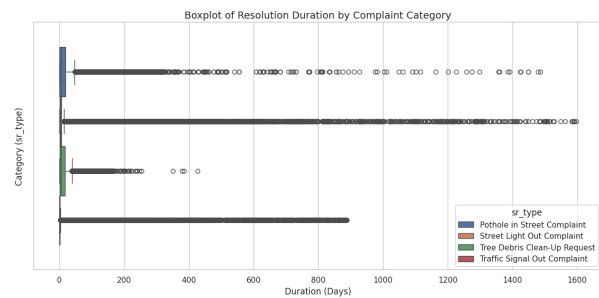


Figure 4. Boxplot of Resolution Duration

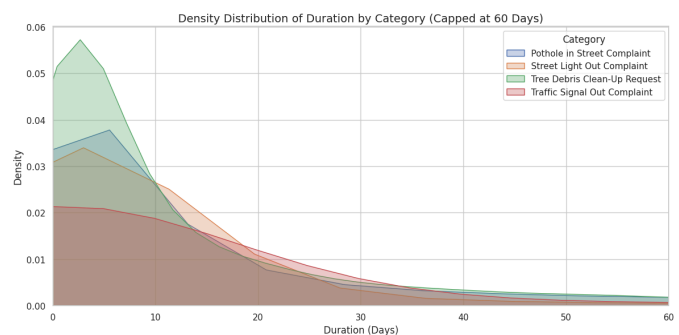


Figure 5. Distribution of Duration

D. Spatial Distribution Graph Construction

Distribution of Infrastructure Complaints with Community Areas (5K Sample)

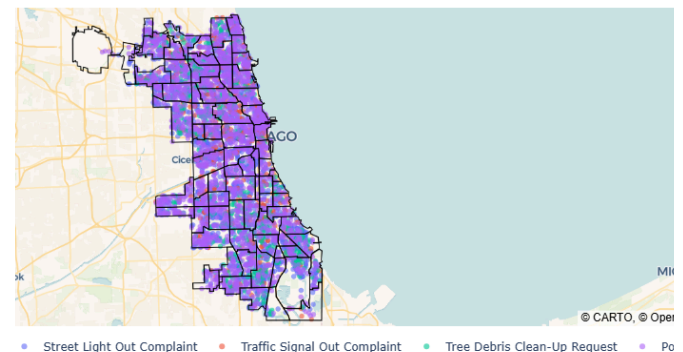


Figure 6. Spatial Distribution of Infrastructure Complaints Across Chicago Community Areas (5K Sample)

A spatial point map was also created using a 5,000-row sample from the cleaned data. This sample was used to keep the visualization readable and easier to render. The map shows the geographic spread of infrastructure complaints across Chicago and overlays the community area boundary layer. This visualization helps confirm that the data

has clear spatial structure and is suitable for graph-based modelling.

For the graph construction, the Chicago community area boundary file was loaded and the areas were sorted by area number. The geometry was projected to EPSG:3435 so that centroid distance could be calculated more accurately. An edge was created when two community areas physically touched each other. The edge weight was calculated using inverse centroid distance, so closer neighbouring areas receive higher weights.

The final spatial graph contains 77 nodes and 394 edges. Each node represents a Chicago community area. This graph is used by the GCN so that the model can learn information from neighbouring areas, instead of treating each area as completely separate.

E. Feature Construction and Data Split

The model input was prepared using three groups of features. The first group consists of spatial node features that represent the characteristics of each community area. These features include the proportion of complaint categories, complaint density, mean duration of completed complaints, ward concentration, electrical district concentration, electricity grid concentration, and average time-related values. These features provide the GCN with information about the infrastructure service conditions of each community area.

The second group consists of temporal features. For each community area, the pipeline constructs a sequence containing daily complaint counts over a 60-day period. This sequence represents recent complaint workload and is used as input to the TCN. The third group consists of incident-level features describing each complaint record, including complaint category indicators, ward, electrical district, creation month, creation hour, creation day of the week, backlog information, and other transformed numerical variables.

The cleaned dataset contains 541,625 records and was divided into four non-overlapping subsets. The training set contains 324,975 records, the validation set contains 108,325 records, the calibration set contains 54,162 records, and the test set contains 54,163 records. This corresponds to a 60/20/10/10 split.

Feature scaling and preprocessing parameters were fitted using training data information to prevent information from the validation, calibration, or test sets from influencing the training process. The same training and test indices were reused for STG-SurviNet, Kaplan-Meier, Cox Proportional Hazards, and Random Survival Forest. Therefore, differences in model performance are not caused by differences in the number of training or testing records. However, each model retains its appropriate feature representation. STG-SurviNet uses the spatial graph, temporal sequences, and incident-level features, while Cox PH and Random Survival Forest use a traditional numerical feature matrix.

F. Initial Model Training Result

The full STG-SurviNet model was successfully trained using the cleaned dataset. The model combines spatial information from the community area graph, temporal workload information from the 60-day complaint history, and incident-level features from each complaint record. The training process was conducted to evaluate whether the proposed architecture could produce meaningful survival predictions and learn patterns from the three feature groups.

The model was trained for up to 50 epochs using a batch size of 32,768, a learning rate of 0.001, a weight decay of 0.0001, and an early stopping patience of 15 epochs. The best validation C-index reached approximately 0.7051 during training. The checkpoint with the strongest validation performance was then

evaluated using the complete test set of 54,163 records.

Table 9. Initial STG-SurviNet Test Evaluation Results

Metric	Test Result	Meaning
Validation loss on test evaluation	9.0751	Loss value reported during test evaluation. Lower is better.
C-index	0.7075	Measures ranking ability. The current result shows moderate ranking performance.
Integrated Brier Score	0.1209	Measures survival probability error over time. Lower is better.
D-Calibration on statistic	8635.4661	Calibration test statistic.
D-Calibration on p-value	0.0000	The survival curves are not well calibrated yet.
MAE	13.68 days	Average absolute error for completed complaints.

The C-index of 0.7075 indicates that STG-SurviNet has a moderate ability to rank complaints according to their resolution risk. This ranking ability is important because the model is expected to distinguish complaints that are more likely to experience longer resolution times. The Integrated Brier Score of 0.1209 indicates a relatively low survival probability prediction error across the evaluated time points.

However, the MAE of 13.68 days shows that the predicted resolution time differs from the actual duration of completed complaints by almost two weeks on average. The D-Calibration p-value of 0.0000 also indicates that the predicted survival probability distributions are not well calibrated. Therefore, the model results are more appropriate for relative risk ranking and pattern analysis than for direct interpretation as absolute survival probabilities.

G. Classical Model Comparison

The implemented classical survival baselines are Kaplan-Meier, Cox Proportional Hazards, and Random Survival Forest. Each model uses the same 324,975 training records and 54,163 test records used in the STG-SurviNet evaluation. No additional record sampling is applied during model fitting or evaluation.

Kaplan-Meier uses the complete training survival outcomes to construct a population-level survival function without covariates. Cox PH and Random Survival Forest use a traditional numerical feature matrix constructed from incident-level features, spatial area features, and temporal workload summaries. Although the feature representations differ according to the requirements of each model, the record-level training and testing splits remain consistent.

Table 10. Classical Model Comparison Result

Model	C-index	IBS	D-Cal p-value	MAE
STG-SurviNet	0.7075	0.1209	0.0000	13.68 days
Cox PH	0.6550	0.1417	0.0000	15.01 days
Random Survival Forest	0.5273	0.1163	0.0000	13.42 days
Kaplan-Meier	0.5000	0.1581	0.0000	15.55 days

The comparison shows that STG-SurviNet achieves the strongest overall performance. Its C-index of 0.7075 is higher than Cox PH at 0.6527, Kaplan-Meier at 0.5000, and Random Survival Forest at 0.4308. This indicates that STG-SurviNet provides the most reliable ranking of complaints according to their predicted resolution risk.

STG-SurviNet also produces the lowest Integrated Brier Score at 0.1209 and the lowest MAE at 13.68 days. Random Survival

Forest produces a lower IBS and MAE than Cox PH, but its C-index of 0.4308 indicates weak risk-ranking performance. Kaplan-Meier has a C-index of 0.5000 because it assigns the same population-level survival pattern to all complaints and does not use complaint-specific covariates. Cox PH is the strongest classical baseline based on its C-index of 0.6527.

All evaluated models produce a D-Calibration p-value of 0.0000. Therefore, none of their predicted survival probability distributions can be considered well calibrated. STG-SurviNet has the lowest D-Calibration statistic among the evaluated models, but the rejected calibration test means that this result should only be interpreted comparatively. Overall, STG-SurviNet is more suitable for ranking complaints and identifying delay-risk patterns than for producing absolute survival probability estimates.



H. Ablation Study Results

The ablation study compares the full STG-SurviNet with three modified versions that remove the spatial component, temporal component, or incident-level features. This directly follows the notebook implementation, where the same trainer class is reused with ablation_mode set to no_spatial, no_temporal, or no_incident. The purpose is to identify

which input group contributes most to survival ranking performance.

Table 11. Ablation Study Result

Model Variant	C-index	IBS	MAE	Change in C-index
Full STG-SurviNet	0.7075	0.1209	13.68 days	0.0000
No Temporal	0.7043	0.1223	13.80 days	-0.0032
No Spatial	0.7029	0.1229	13.82 days	-0.0046
No Incident Features	0.5467	0.1474	15.14 days	-0.1608

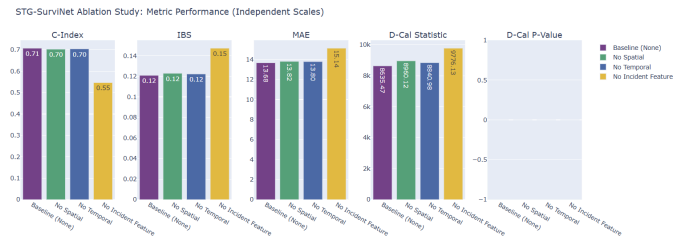


Figure 7. STG-SurvivalNet Ablation Study: Metric Performance (Independent Scale)

The ablation result shows that removing incident-level features causes the largest performance drop, reducing the C-index by 0.1608 compared with the full model. Removing the temporal component reduces the C-index by only 0.0032, while removing the spatial component reduces it by 0.0046. This indicates that incident-level information is the strongest contributor in the current implementation, while the GCN and TCN components still add smaller but measurable value.

I. Spatial Embedding Clustering

The trained GCN embeddings are clustered using a Gaussian Mixture Model to identify groups of community areas with similar

learned spatial representations. The number of clusters is selected from k values between 2 and 8 using BIC. The lowest BIC is obtained at k equal to 2, so the spatial analysis uses two GMM clusters.

Table 12. Spatial Embedding Clustering Result

Cluster	Records	Mean Duration	Median Duration	Event Rate
0	389,266	19.31 days	3 days	0.9724
1	152,359	18.35 days	3 days	0.9678

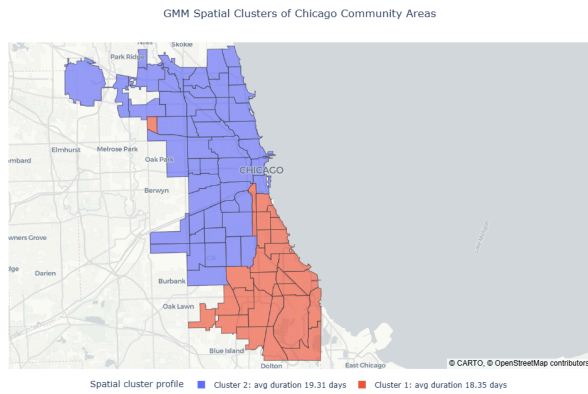


Figure 8. GMM Spatial Clusters of Chicago Community Areas

The two spatial clusters have close median durations of 3 days, but their mean durations and event rates differ slightly. Cluster 0 contains 389,266 records with a mean duration of 19.31 days, while Cluster 1 contains 152,359 records with a mean duration of 18.35 days. Since the cluster labels are generated by GMM, the labels are interpreted as group identifiers rather than ordered risk levels.

J. Temporal Workload Pattern Analysis

The temporal analysis uses the 60-day workload sequence that is already constructed for the TCN input. GMM is also applied to the TCN embeddings, and the best BIC is obtained at k equal to 3. This creates three temporal workload groups that summarize how

community areas differ in recent complaint volume.

Table 13. Temporal Workload Pattern Analysis Result

Cluster	Number of Areas	Mean 60-Day Workload	Mean Daily Workload	Mean Peak Daily Workload
2	10	326.90	5.45	19.60
0	22	247.59	4.13	17.09
1	45	142.82	2.38	12.87

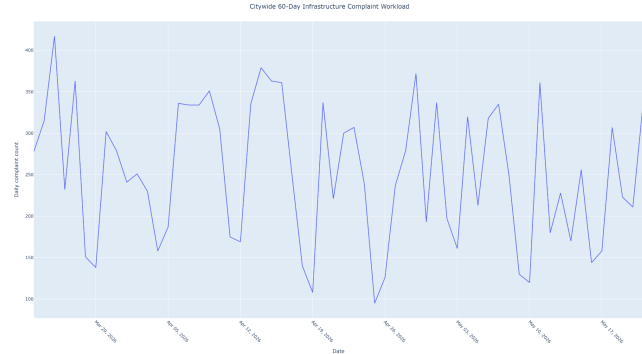


Figure 9. Citywide 60-Day Infrastructure Complaint Workload

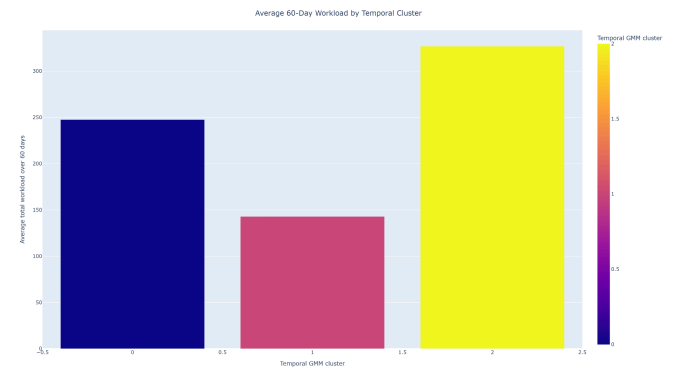


Figure 10. Average 60-Day Workload by Temporal

The temporal clusters separate high, medium, and lower workload areas based on recent complaint volume. Cluster 2 contains only 10 areas but has the largest mean 60-day workload at 326.90 complaints. The highest workload areas include Austin with 509 complaints and Near West Side with 493 complaints during the 60-day window. These results support the use of temporal features

because workload varies substantially across community areas.

K. Delay Risk Pattern Analysis

Delay risk is calculated from the log hazard output produced by the full STG-SurviNet model. The variable `delay_risk_score` is defined as the negative log hazard. Therefore, a higher delay-risk score represents a lower predicted resolution hazard and a greater predicted risk of prolonged complaint resolution. The score is aggregated by complaint category and community area to summarize the patterns learned by the model.

Table 14. Delay Risk Pattern Analysis Result

Complaint Category	Records	Mean Duration	Median Duration	Mean Delay-Risk
Tree Debris Clean-Up Request	60,384	13.85 days	5 days	0.2005
Pothole in Street Complaint	243,487	22.43 days	5 days	0.0186
Street Light Out Complaint	188,897	15.68 days	2 days	-0.2680
Traffic Signal Out Complaint	48,857	21.54 days	0 days	-0.5163

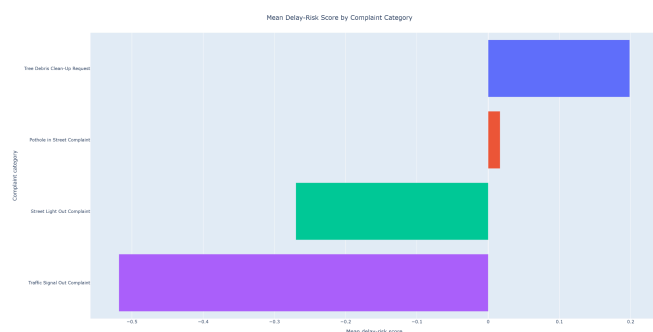


Figure 11. Mean Delay-Risk Score by Complaint Category

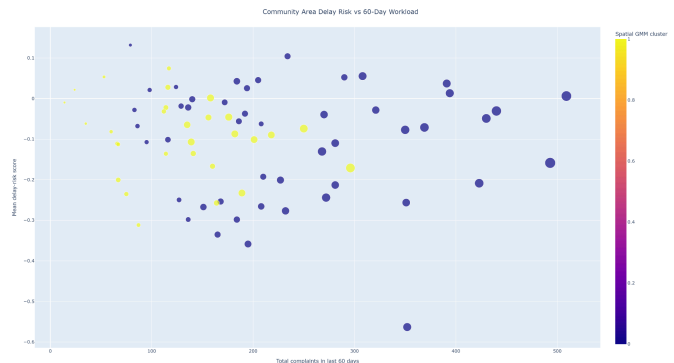


Figure 12. Community Area Delay Risk vs 60-Day Workload

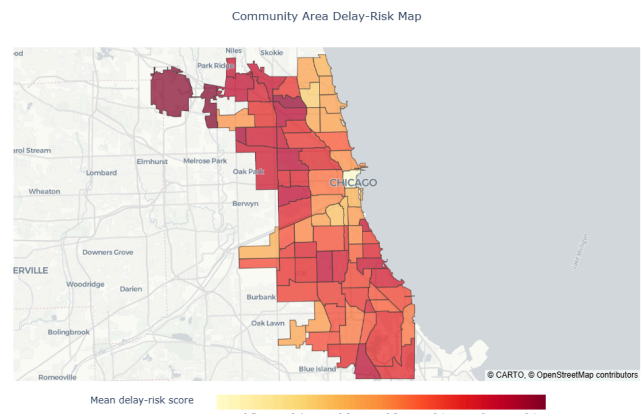


Figure 13. Community Area Delay-Risk Map

Tree Debris Clean-Up Request has the highest mean delay-risk score at 0.1985, followed by Pothole in Street Complaint at 0.0167. Street Light Out Complaint and Traffic Signal Out Complaint produce lower average scores at -0.2699 and -0.5183, respectively. These values represent relative model outputs, so a negative value does not indicate that a complaint has no delay. It indicates a lower predicted delay risk relative to categories with higher scores.

At the community-area level, O'Hare has the highest mean delay-risk score at 0.1319. It contains 1,834 records, has a mean resolution duration of 29.95 days, and has a median duration of 6 days. At the area-category level, the highest delay-risk combination is Roseland with Tree Debris Clean-Up Request. This combination contains 2,139 records, has a mean duration of 22.79 days, a median

duration of 11 days, and a mean delay-risk score of 0.5497.

These findings describe associations learned by STG-SurviNet from the available spatial, temporal, and incident-level information. They should be interpreted as model-based risk patterns rather than causal explanations of complaint resolution delays.

L. Remaining Model Limitation

While the analysis demonstrates the potential of STG-SurviNet as a ranking and pattern-discovery tool, the model still has several limitations. Its test C-index of 0.7075 remains below the target of 0.75, while its MAE of 13.68 days remains above the target of 7 days. All evaluated models also produce a D-Calibration p-value of 0.0000, indicating that their predicted survival probability distributions are not well calibrated. Although STG-SurviNet achieves the strongest C-index, Integrated Brier Score, and MAE, its probability estimates should not be interpreted as reliable absolute predictions.

Random Survival Forest produces a C-index of 0.4308, indicating weak ranking performance under the implemented features and parameters, but this result should not be generalized to every Random Survival Forest implementation. Furthermore, the identified spatial, temporal, and delay-risk patterns represent associations learned from the Chicago 311 data and do not establish causal relationships. Therefore, the findings are interpreted as comparative ranking and model-based pattern evidence rather than causal conclusions.

Chapter 4 Conclusion and Recommendations

A. Conclusion

This study implemented STG-SurviNet, a hybrid survival analysis architecture combining a Graph Convolutional Network, Temporal Convolutional Network, and Deep

Cox Proportional Hazards layer to model infrastructure complaint resolution times across Chicago's 77 community areas. The model was evaluated against Kaplan-Meier, Cox Proportional Hazards, and Random Survival Forest using the same 324,975 training records and 54,163 test records. STG-SurviNet achieved the highest C-index of 0.7075, outperforming Cox PH at 0.6527, Kaplan-Meier at 0.5000, and Random Survival Forest at 0.4308. It also produced the lowest Integrated Brier Score of 0.1209 and the lowest MAE of 13.68 days. These results show that combining spatial, temporal, and incident-level information provides stronger ranking performance and lower prediction error than the evaluated classical baselines.

The ablation study shows that incident-level features make the largest contribution to model performance, as removing them reduces the C-index by 0.1608. Removing the spatial component reduces the C-index by 0.0046, while removing the temporal component reduces it by 0.0032. The spatial and temporal components therefore provide smaller but measurable contributions to the complete architecture. The pattern analysis identifies Tree Debris Clean-Up Request as the complaint category with the highest average delay-risk score and O'Hare as the community area with the highest average delay-risk score. Temporal clustering also separates the 77 community areas into three workload groups, with Austin and Near West Side recording the highest complaint volumes during the analyzed 60-day window.

Despite these results, all evaluated models produce a D-Calibration p-value of 0.0000. STG-SurviNet also remains below the original C-index target of 0.75, while its MAE of 13.68 days remains above the target of 7 days. Therefore, the model should be interpreted primarily as a risk-ranking and

pattern-discovery tool rather than a fully calibrated duration prediction system. The identified delay-risk patterns provide evidence of model-learned associations, but they do not establish causal explanations for differences in complaint resolution times.

B. Recommendations

Based on the findings of this study, city infrastructure management could direct additional monitoring toward complaint categories and community areas with consistently high predicted delay risk. Tree Debris Clean-Up Request has the highest category-level delay-risk score, while O'Hare has the highest community-area score. Roseland combined with Tree Debris Clean-Up Request produces the highest area-category delay-risk score at 0.5497. These areas and complaint categories could be reviewed when allocating field resources, although the model results should be combined with operational information before decisions are made. The ten community areas in Temporal Cluster 2, including Austin and Near West Side, could also be monitored through their 60-day complaint workload because this cluster records the highest average workload.

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Appendix

Appendix A

The code and files related to this analysis are available at the following Google Colab link:

[!\[\]\(758ebdf4629c903da74c2e079717ae32_img.jpg\) DatMin.ipynb](#)

Appendix B

The files related to this analysis are available at the following Google Drive link:

[!\[\]\(a8f9309f944226d1420f5fed22e2b6e6_img.jpg\) Progress Group 4](#)